



15
30

Numerical Methods

Mid Term Exam

Mark:

Student's Name:

ID Number:

Date of Exam: 27-11-2022 Sunday (14:30-15:20)

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Question1: Solve the first two iterations to find the root of $f(x) = -1 + 5.5x - 4x^2 + 0.5x^3$ and the $x_0 = 0.25$ using Newton Raphson Method.

$$f'(x) = 5.5 - 8x + 1.5x^2$$

n	x_n	x_{n+1}
1	0.25	0.21304
2	0.21304	0.2143314051

3

Question2: Based on the bisection method, find the root of $f(x) = 3x^2 - e^x$ that is located in $[3.5, 4]$ with error less than $\epsilon = 0.1$

n	x_n	x_c	$f(x_n)$	$f(x_c)$	x_r	$f(x_r)$	ϵ
1	3.5	4	3.63454	-6.598	3.75	-0.3335	—
2	3.5	3.75	3.634	-0.3335	3.625	1.8971	3.997 0.125
3	3.625	3.75	1.8971	-0.3335	3.6875	0.8481	0.0625

ans
Question 3: Use Gauss elimination to find a where $x_1 = 3$

$$30ax_1 - ax_2 - 2ax_3 = 7.8$$

$$ax_1 + 70ax_2 - 3ax_3 = -19.3$$

$$3ax_1 - 2ax_2 + 100ax_3 = 71.4$$

$$\begin{bmatrix} 30a & -a & -2a \\ a & 70a & -3a \\ 3a & -2a & 100a \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7.8 \\ -19.3 \\ 71.4 \end{bmatrix}$$

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Question4: Find the value β that the system

$$\begin{bmatrix} 1 & 1 & -1 \\ 2 & \beta & 1 \\ 3 & 2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \\ -4 \end{bmatrix} \text{ so}$$

that no inverse exists (Singular).

~~$\det = 0 \Rightarrow 1(-4\beta - 2) - 1(-8 - 3) + 1(4 - 2\beta) = 0$~~

~~$\Rightarrow -4\beta - 2 - (-8 - 3) + 4 - 2\beta = 0$~~

$1(-4\beta - 2) - 1(-8 - 3) - 1(4 - 3\beta) = 0$

$\beta = 5$

2200

Question 5: if $Ax = b$

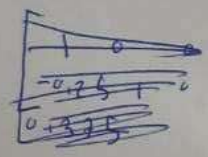
$[L] = \begin{bmatrix} 1 & 0 & 0 \\ -0.25 & 1 & 0 \\ 0.375 & 0.23913 & 1 \end{bmatrix}$, and $[U] = \begin{bmatrix} -8 & 1 & -2 \\ 0 & -5.75 & -1.5 \\ 0 & 0 & 8.108696 \end{bmatrix}$,

If $[b]^T = [-20 \ -38 \ -34]$, find: (show details of solution)

1. $[A]$
2. x_3 ?

①

$[L] \cdot [U] = [A] \Rightarrow [A] = \begin{bmatrix} -8 & 1 & -2 \\ 3 & -5.75 & 0 \\ -7.76087 & 8.108696 & -2.8749975 \end{bmatrix}$



$\begin{bmatrix} -8 & 1 & -2 \\ 3 & -5.75 & 0 \\ -7.76087 & 8.108696 & -2.8749975 \end{bmatrix} \begin{bmatrix} -20 \\ -38 \\ -34 \end{bmatrix}$

~~$[A] \cdot [x] = [b]$~~

~~$\begin{bmatrix} -8 & 1 & -2 \\ 3 & -5.75 & 0 \\ -7.76087 & 8.108696 & -2.8749975 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -20 \\ -38 \\ -34 \end{bmatrix}$~~

②

~~$[U] \cdot [x] = [b]$~~

$a_{33} x_3 = d_3$

$(8.108696) x_3 = -34$

$x_3 = -4.193029361$